MiniPic: Benchtop Vacuum Tweezers
Assembly Manual

Version: 1.0

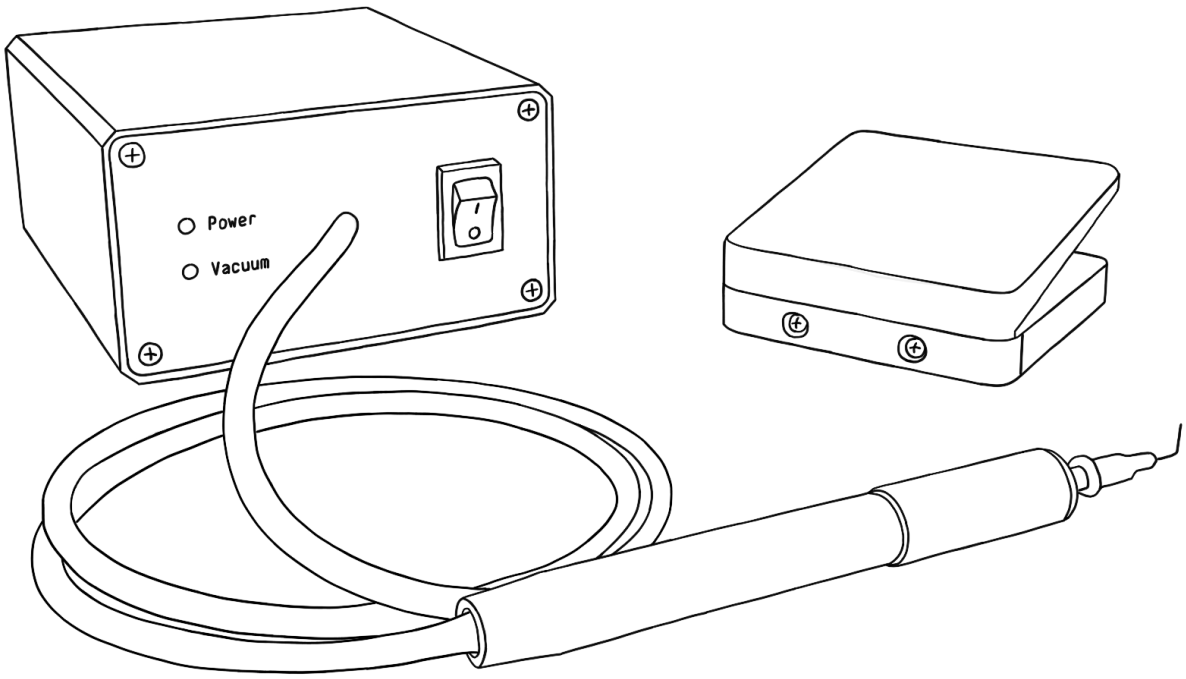


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Quickstart Guide

The foot pedal can be configured to be normally open (NO), meaning that the vacuum only activates when the pedal is pressed. If the device is switched into normally closed (NC) mode, then the vacuum will be on unless interrupted by the foot pedal. Feel free to try both modes and find what works best for you.

There are two LEDs on the front panel of the MiniPic. The green “Power” LED lights up whenever USB-C power and the power switch is turned on. The blue “Vacuum” LED lights up whenever the vacuum motor and solenoid are activated.

All source files, 3D models, and a bill of materials can be found on GitHub at: https://github.com/ARENARobotics/MiniPic

What's in the Box

Main Unit

➔ Main PCB	(x1)	➔ Screws	(x25)
➔ LED Holder	(x1)	➔ 20mm Tube	(x1)
➔ LED Cable	(x1)	➔ 40mm Tube	(x1)
➔ Power Switch	(x1)	➔ 3m Tube	(x1)
➔ Power Cable	(x1)	➔ Front Plate	(x1)
➔ Vacuum Motor	(x1)	➔ Back Plate	(x1)
➔ Solenoid	(x1)	➔ Base	(x1)
➔ Filter	(x1)	➔ Lid	(x1)
➔ Rubber Foot	(x4)	➔ Vacuum Cover	(x1)
➔ Pen Body	(x1)	➔ Solenoid Cover	(x1)
➔ Box of Pen Tips	(x1)	➔ Filter Cover	(x1)
➔ USB-C Cable	(x1)		

Foot Pedal

➔ Pedal PCB	(x1)	➔ Screws	(x14)
➔ Key Switch	(x1)	➔ Pedal Base	(x1)
➔ Hinge	(x1)	➔ Pedal Top	(x1)
➔ Aux Cord	(x1)	➔ Front Bumper	(x1)
➔ Rubber Foot	(x4)		

Spare Parts

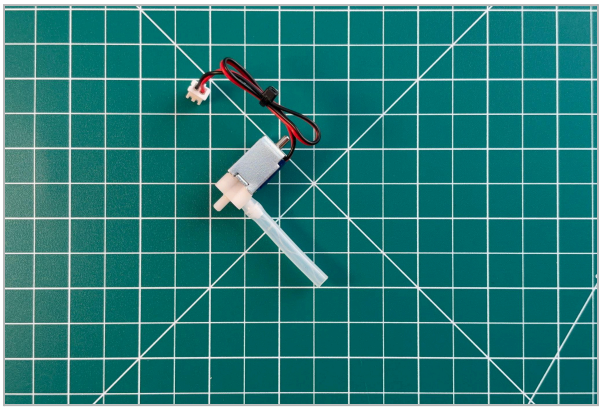
➔ Screws	(x3)

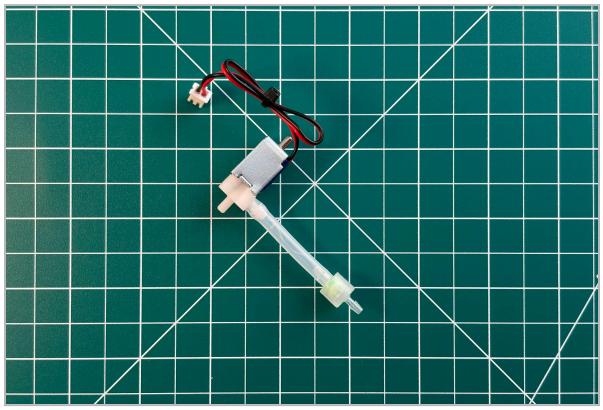
Main Unit Assembly:

Note

ⓘ Parts shown in this manual may appear differently than what was shipped. Any differences are cosmetic, or in any case will not affect the assembly of the device.

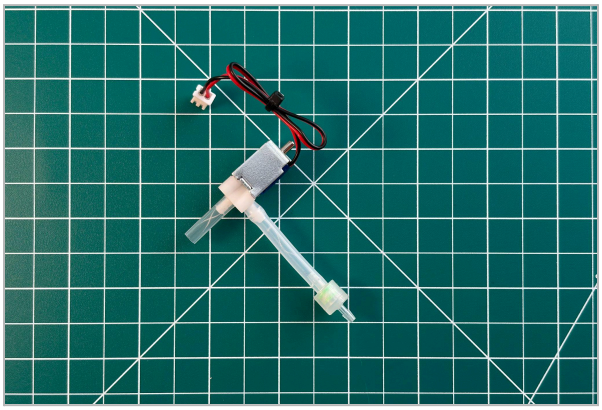
STEP 1: Plug 40mm Tube onto Solenoid and Filter

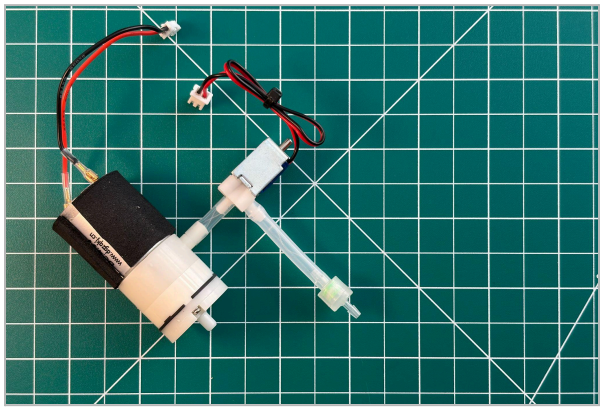




- ➔ 40mm Tube must be connected to the side port of the Solenoid
- ➔ 40mm Tube can connect to either port of the Filter

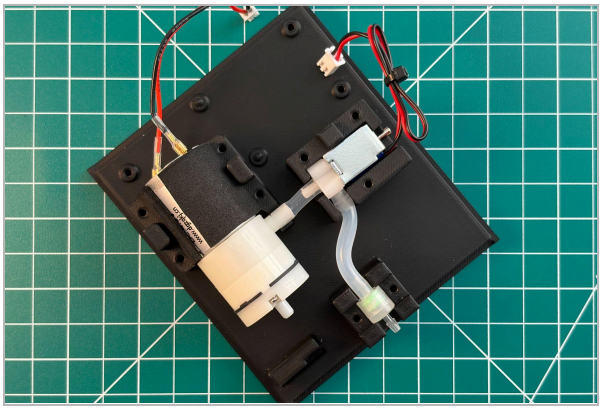
STEP 2: Plug 20mm Tube onto Solenoid and Vacuum Motor

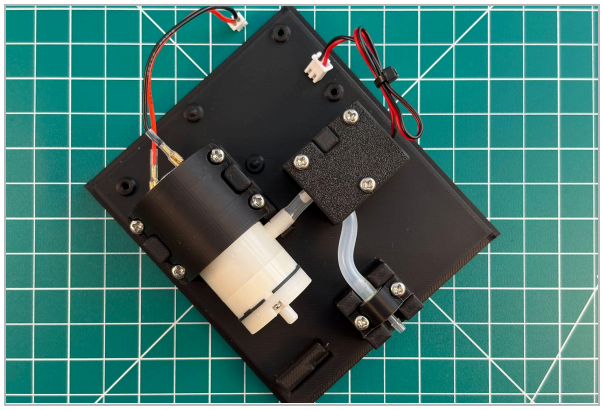




- ➔ 20mm Tube must be connected to the Solenoid port opposite its cable
- ➔ 20mm Tube must be connected to the side port of the Vacuum Motor
- ➔ 20mm Tube must be fully compressed against the ends of the ports on the Solenoid and Vacuum Motor

STEP 3: Install Solenoid, Vacuum Motor, and Filter

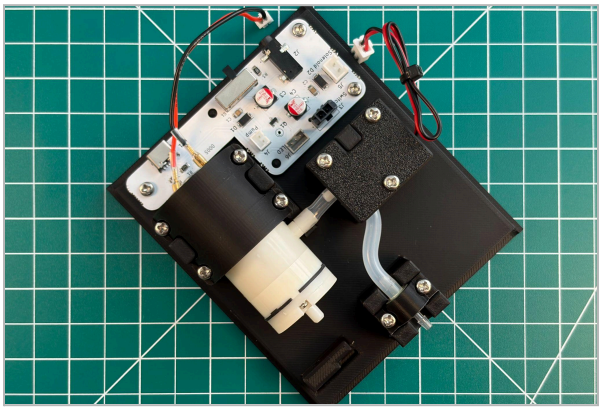




➔ Screw in all covers using included screws

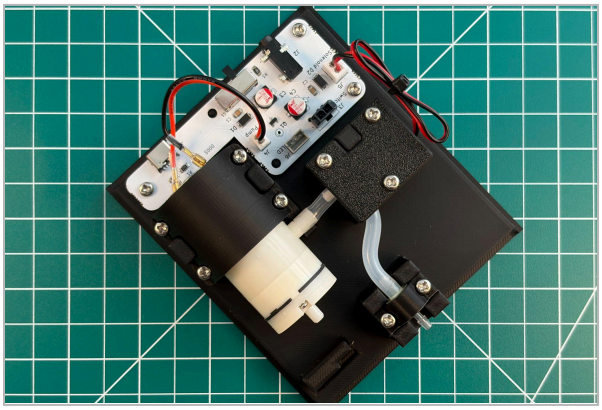
⚠ A 1mm gap should be maintained between the Vacuum Cover and the Base so that the foam on the vacuum motor is not over compressed

STEP 4: Install Main PCB



➔ Screw in using included screws

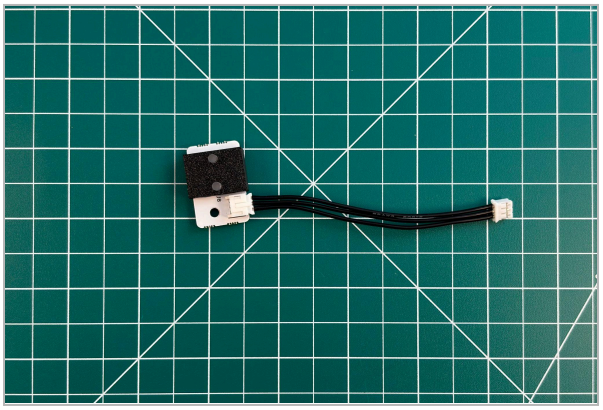
STEP 5: Plug in Vacuum Motor and Solenoid

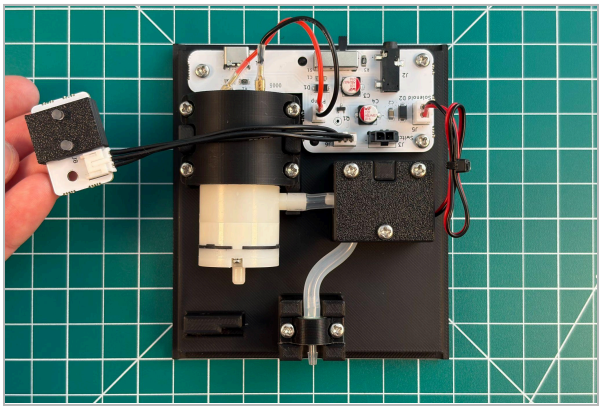


➔ Plug Solenoid into Main PCB
➔ Plug Vacuum Motor into Main PCB

ⓘ Note orientation of plugs, each can only be inserted in one orientation

STEP 6: Attach LED Cable

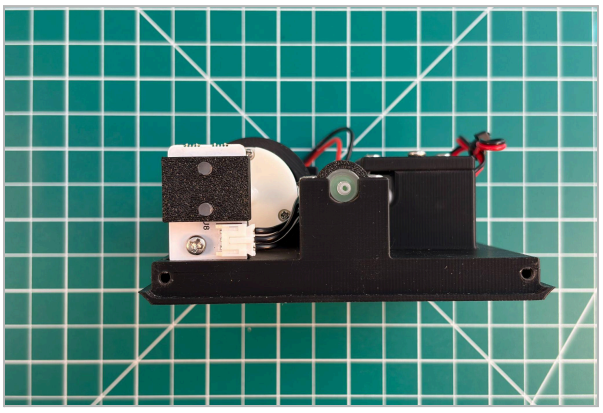




➔ Plug LED Cable into LED Holder, then Main PCB

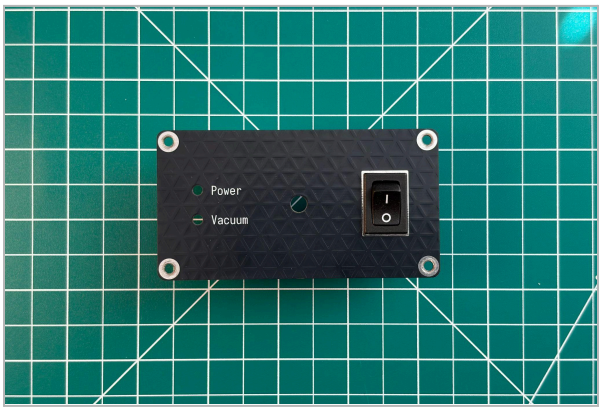
ⓘ Note orientation of plugs, each can only be inserted in one orientation

STEP 7: Install LED Holder



➔ PCB must be fully seated in the Base

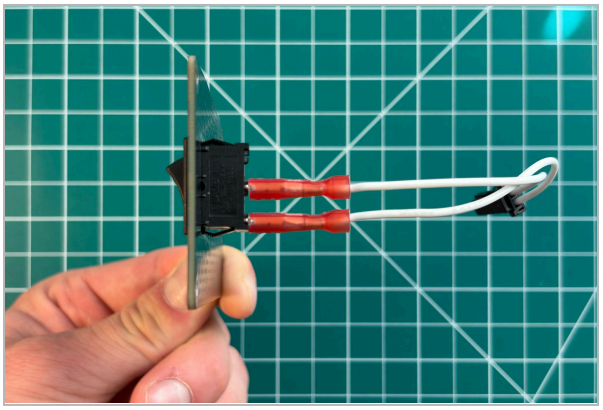
STEP 8: Install Power Switch

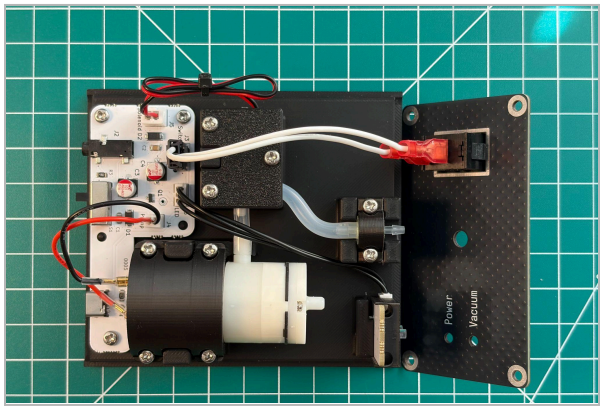


➔ Snap Power Switch into Front Plate until the border of the switch is flush against the Front Plate

➔ Note orientation of the Power Switch relative to the Front Plate

STEP 9: Attach Power Cable



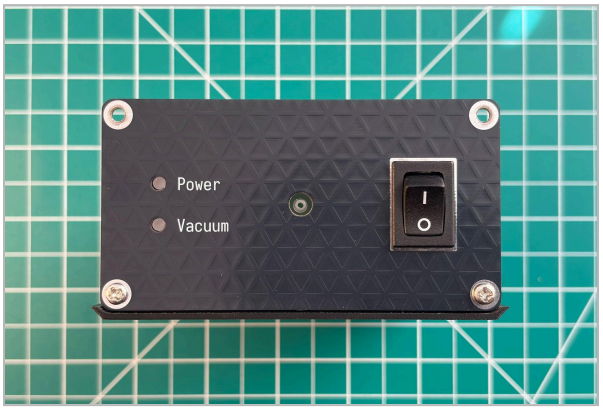


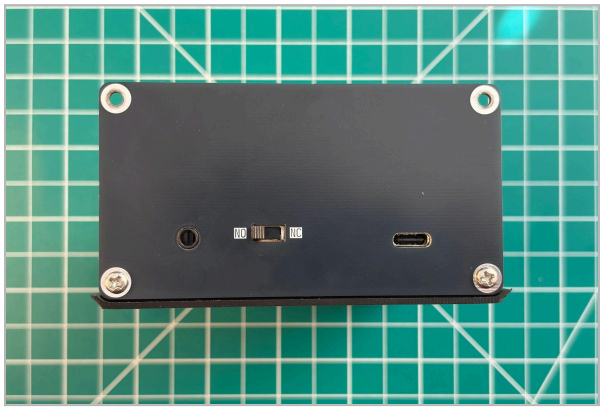
➔ Plug red contacts onto the terminals on the Power Switch. Orientation and placement do not matter.

➔ Plug the connector on the opposite end of the cable into the Main PCB

ⓘ Note orientation of the connector that plugs into the Main PCB. The connector can only be inserted in one orientation.

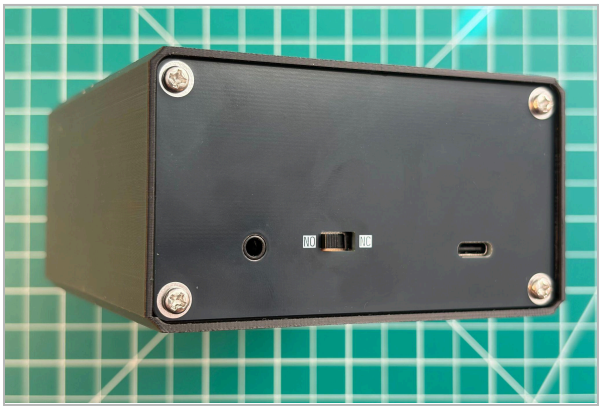
STEP 10: Install Front and Back Panels





➔ Screw in Front Panel and Back Panel with included screws

STEP 11: Install Lid

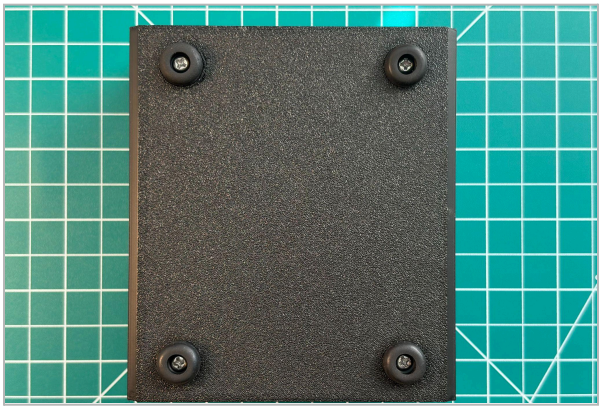




➔ Lower Lid onto Base, and then install using included screws

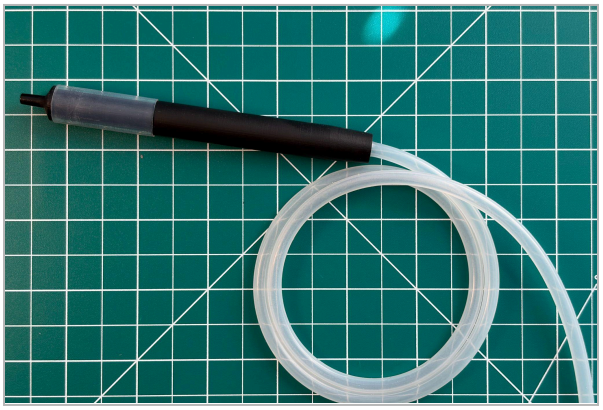
⚠ Ensure that wires are not pinched when closing the lid! Pinched cables can get damaged and cause unexpected behavior!

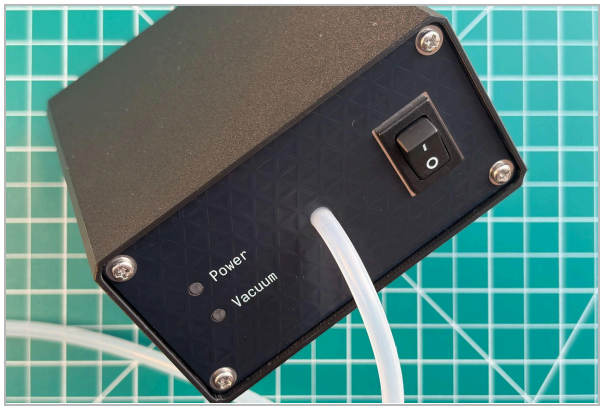
STEP 12: Install feet



➔ Screw in feet using included screws

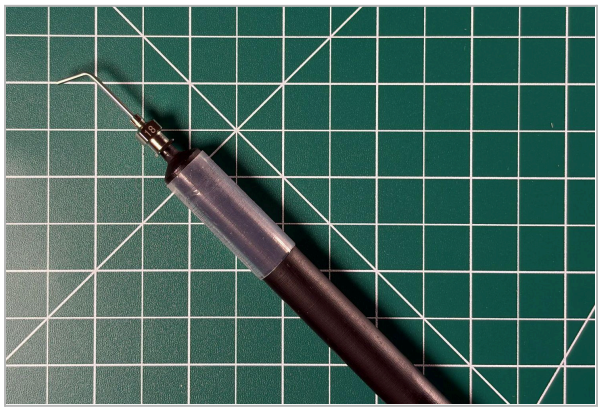
STEP 13: Attach Pen

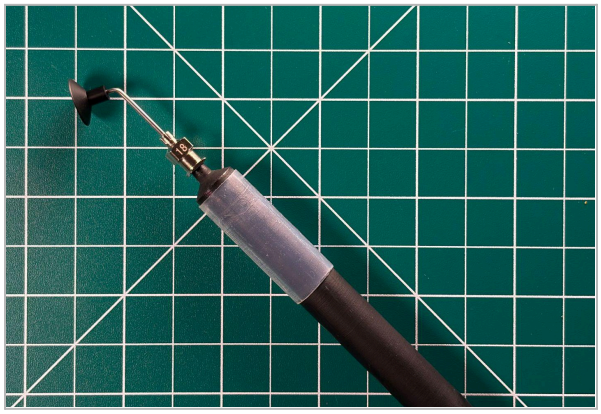




➔ Plug 3m Tube into Pen Body, then into Main Unit

STEP 14: Attach Pen Tip





➔ Press a Pen Tip onto end of Pen Body

➔ A variety of needle tips are included for use with different size components. Larger needle gauges generally work better for larger components.

➔ Multiple suction cups are included for use with heavy components. To use a suction cup, simply press one onto the end of a needle tip. The suctions cups work best on the 18 gauge all metal tip and the 18 gauge green tip.

⚠ Plastic pen tips can crack over time if too much force is used to press them onto the pen body. We recommend inserting them lightly, and then pressing harder if the vacuum is too weak.

STEP 15: Test out the Main Unit!

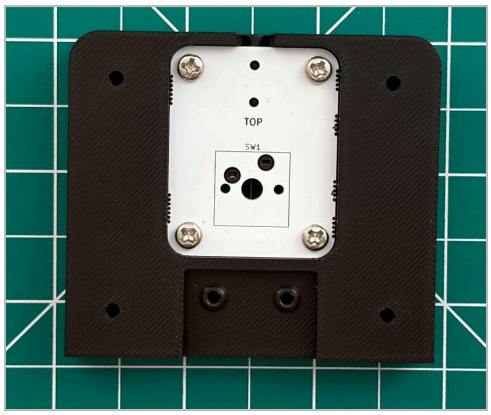
Once plugged into a USB power source, the pen can be operated using just the power switch on the front or the slide switch on the back. The vacuum pump should turn on once the device is powered on. If no suction can be felt at the pen tip, review all tubing and electrical connections.

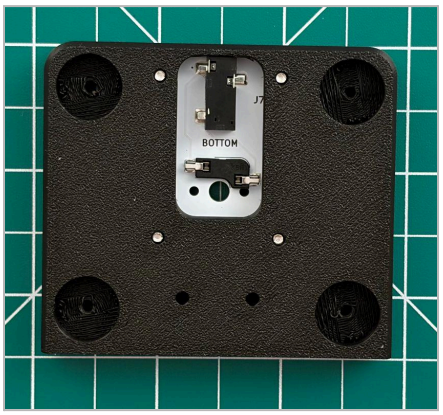
Foot Pedal Assembly

Note

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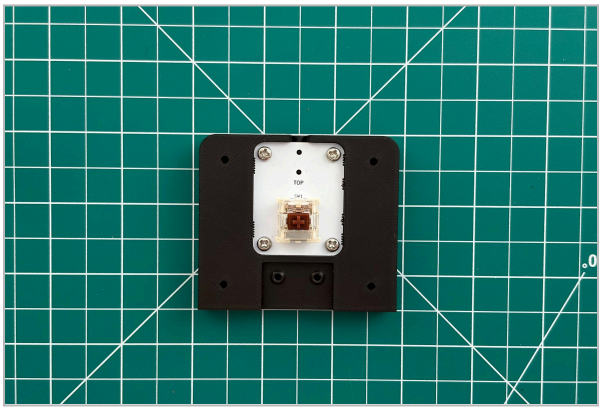
STEP 1: Install Pedal PCB





➔ Note: the labels on the Pedal PCB should go on the corresponding face of the Pedal Base. Feet screw into the bottom face orientation of Pedal Base.

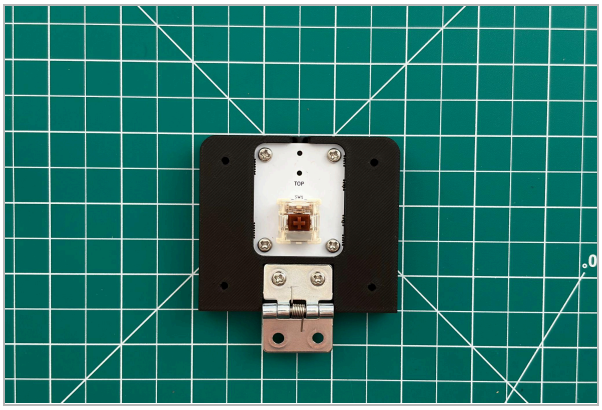
STEP 2: Insert Key Switch

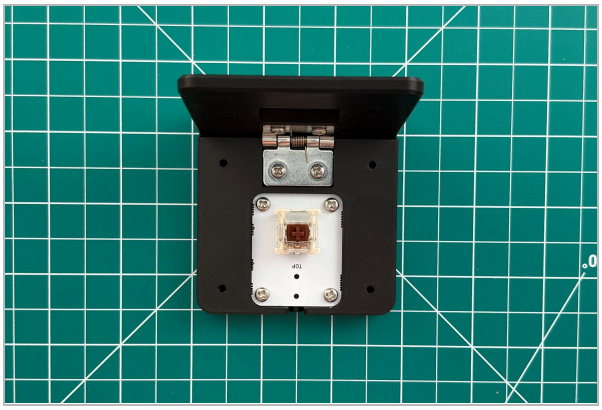


➔ Press Key Switch into Pedal PCB

⚠ Use alignment posts on Key Switch to ensure the metal terminals do not bend

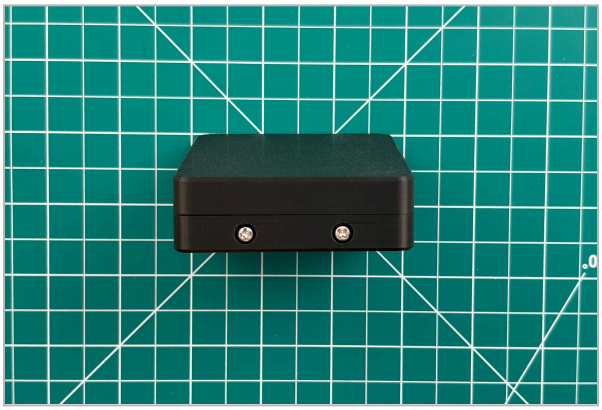
STEP 3: Install Hinge





➔ Screw into Pedal Base and then Pedal Top

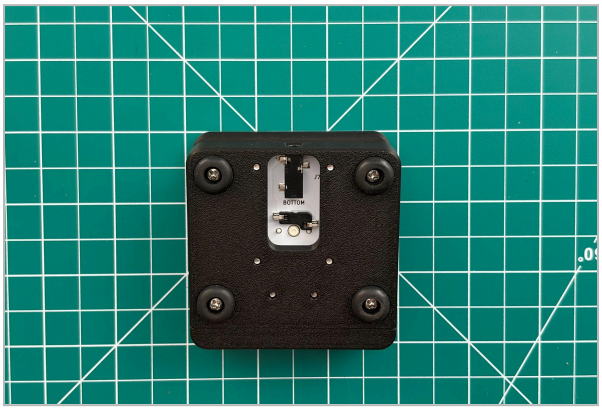
STEP 4: Install Front Bumper



➔ Pedal Top must be pressed against the Pedal Base

➔ Screws installed in this step can strip easily, so tighten slowly

STEP 5: Install Feet



➔ Install using included screws

STEP 6: Plug in Aux Cord





➔ Insert each connector until almost no brass is visible on the connector

STEP 7: Start Building!

With the foot pedal attached, your MiniPic is now fully assembled! The foot pedal can be configured to be normally open (NO), meaning that the vacuum only activates when the pedal is pressed. If the device is switched into normally closed (NC) mode, then the vacuum will be on unless interrupted by the foot pedal. Feel free to try both modes and find what works best for you.